

STA630

Final-Term (Solved)

ABSTRACT

This comprehensive collection of notes is accurately crafted to empower students to excel academically, ensuring they achieve a minimum of 80% marks in their examinations. The content is organized with clarity and precision, focusing on key concepts, critical analyses, and practical applications tailored to the syllabus. These notes serve as a reliable resource for both thorough preparation and last-minute revision. Designed to inspire confidence and mastery, this guide is an essential tool for students striving for academic excellence.

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RESEARCH METHODS

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Experimental research
Experimental research is a systematic method where researchers manipulate one variable and control others to observe effects. It involves hypothesis testing and control groups to ensure internal validity.

Quasi-experimental designs
These are similar to true experiments but lack random assignment. Common examples include pretest-posttest designs without randomization.

True experimental designs
These include random assignment, control and experimental groups, and manipulation of independent variables. This helps establish cause-and-effect relationships.

One-shot case study design
This design involves exposing a single group to a treatment and then measuring the outcome. It lacks comparison or control, making internal validity weak.

Survey research
This involves collecting data through questionnaires or interviews to describe characteristics, opinions, or behaviors of a population.

Historical-comparative research
This method compares historical events or social structures across time or cultures. It's useful in social sciences to understand changes and differences.

Non-reactive research (with examples)
This method collects data without influencing the subject, e.g., content analysis, use of existing statistics, or observation from a distance.

Field research
Involves collecting data in a natural environment, using participant observation or unstructured interviews. It aims to understand phenomena in context.

Quantitative approach
Relies on numerical data, statistical analysis, and objective measurements. Used for hypothesis testing and generalizations.

Qualitative approach
Focuses on exploring phenomena through text, observation, or interviews. It aims for depth of understanding rather than measurement.

Longitudinal and cross-sectional study
Longitudinal studies collect data over a period of time to detect changes, while cross-sectional studies gather data at one point in time.

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Identify time dimension of study (longitudinal or cross-sectional)
If data is collected at different points (e.g., two months or years), it is longitudinal. If data is collected at a single point, it is cross-sectional.

Scenario-based identification of research types (mobile features vs. automobile survey)

- Mobile feature scenario: **Descriptive research**
- Automobile survey assessing methods: **Evaluative research**

Economic study of Pakistan – suitable research type?
For studying economics, **quantitative research** using **secondary data analysis** or **survey methods** is commonly appropriate.

Observational study type in a cultural setting
This is a **participant observation**, typical in **qualitative field research** where researchers immerse in the setting to understand behaviors.

Field and its significance in field research
The field refers to the natural setting where the subject exists. Studying in the field enhances realism and context-based understanding.

Similarities to field research
Like qualitative research, field research involves real-world settings, open-ended data collection, and interpretive analysis.

Sampling – definition and importance
Sampling is selecting a subset of individuals from a population to make generalizations. It saves time, cost, and allows manageability.

Why researcher takes sample instead of whole population
Due to time, cost, and logistical constraints, sampling provides a practical and efficient way to study large populations.

Probability and non-probability sampling (comparison)

- **Probability sampling**: each unit has a known, equal chance (e.g., random sampling) – more generalizable.
- **Non-probability sampling**: selection is non-random (e.g., convenience sampling) – less generalizable but easier.

Simple random sampling
Each member of the population has an equal chance of being selected using random methods (e.g., lottery method).

Systematic random sampling
Every kth individual is selected from a list after a random start.

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Stratified random sampling
Population is divided into strata (e.g., gender, age) and random samples are taken from each stratum.

Cluster sampling
Entire clusters (e.g., schools, regions) are randomly selected and all individuals within are studied.

Snowball sampling
Used in hidden or hard-to-reach populations; participants refer other participants.

Quota sampling
Researcher ensures specific characteristics are represented in the sample (e.g., gender ratio).

Purposive sampling
Subjects are selected based on specific purpose or characteristics relevant to the study.

Convenience sampling
Sample is drawn from those easiest to access.

Sequential sampling
Data is collected and analyzed as it is being gathered, and sampling continues until enough data is acquired.

Theoretical sampling
Used in qualitative research; sampling continues as theory develops.

Sampling frame and sampling frame error
Sampling frame is the list from which the sample is drawn. Errors occur if the frame does not accurately represent the population.

Disproportionate and proportionate sampling

- **Proportionate**: sample reflects the population proportions.
- **Disproportionate**: deliberately oversamples smaller groups.

Sample population – why used
To represent the characteristics of a larger population for efficient and cost-effective research.

Scenario: researcher selects 15 people, then 5
This is **purposive sampling** or could be **sequential sampling** depending on the context.

ERP employees easy to access – sampling type
This is **convenience sampling**, a type of **non-probability sampling**.

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Similar population – why preferred
Homogeneity reduces variability and increases internal validity, especially in experimental designs.

Pretest and posttest
Measurements taken before and after the treatment to assess change due to the intervention.

Control group
Group that does not receive the experimental treatment, used as a baseline for comparison.

Random assignment
Assigning participants to groups by chance to ensure each group is similar and controls bias.

Double-blind experiment
Neither participants nor researchers know who is in the control or experimental group, reducing bias.

Instrument effect in internal validity
Refers to changes in measurement tools or procedures that can affect outcomes, threatening internal validity.

Explain role of pre-test and post-test with example
A pre-test gauges initial ability; a post-test measures changes after treatment.
Example: Testing student scores before and after a new teaching method.

Maturation effect and test effect

- **Maturation**: Natural changes over time that affect results (e.g., fatigue, aging).
- **Test effect**: Taking the test more than once influences results due to familiarity.

Maturation effect on internal validity with example
Maturation refers to natural changes in subjects over time (e.g., fatigue, aging) that can influence results. For example, students may perform better post-test not due to teaching but due to natural cognitive development over weeks.

Internal and external validity – difference
Internal validity measures if the change in the dependent variable is caused by the independent variable. External validity refers to how generalizable the results are to other settings or populations.

Factors affecting internal validity
Common factors include maturation, testing, instrumentation, selection bias, and experimental mortality.

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Importance of instrumentation in internal validity
Changes or inconsistencies in the tools used for measurement can threaten internal validity. Standardized and reliable instruments help maintain validity.

Constructing frequency distribution for two variables
This involves organizing data into a bivariate table where each cell shows the frequency of occurrence for the combination of two variables.

Bivariate tables
These tables display the relationship between two variables, usually in rows and columns.

Trivariate tables – two limitations
They become complex to interpret and may mask simpler bivariate relationships. There's also a higher risk of misinterpretation due to interaction effects.

Central tendency and its three methods
Measures of central tendency summarize a data set with a single value: Mean (average), Median (middle value), and Mode (most frequent value).

Index construction
An index combines multiple indicators into a single measure. Construction involves item selection, scaling, and scoring rules.

Parameter and statistics – difference
A parameter refers to a value describing a characteristic of a population, while a statistic describes a characteristic of a sample.

Statistics and population
Statistics are calculated from sample data and used to estimate population parameters.

Data transformation with example
Data transformation involves converting data into another format for analysis (e.g., converting income ranges into numerical scores for comparison).

Coding – two basic rules (mutually exclusive, exhaustive)
Codes should be mutually exclusive (each response fits one category only) and exhaustive (all possible responses are covered).

Basics of coding construction
Involves creating a codebook, assigning numeric or symbolic codes to responses, and ensuring consistency during data entry and analysis.

Questionnaire survey
A structured method for collecting data from respondents using a set of written questions.

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Pilot testing – importance and steps
Pilot testing helps identify flaws in the research instrument. Steps include administering the tool to a small sample, analyzing feedback, and revising accordingly.

Pilot testing of the questionnaire
This pre-testing process helps ensure clarity, reliability, and validity of the questionnaire items before the full study.

What aspects to evaluate during pilot testing
Question clarity, wording, format, response time, and consistency of responses should be assessed.

Guidelines for questionnaire
Use simple, clear language; avoid leading or double-barreled questions; maintain logical flow; and ensure relevance to objectives.

Cover letter – what it should include
It should explain the study purpose, ensure confidentiality, provide contact information, and request voluntary participation.

Leading, loaded, double-barreled, burdensome questions – identification
Example: “I am satisfied with the salary and the work environment” is double-barreled. Each type of flawed question biases or confuses responses.

Editing – what and why
Editing ensures completeness, accuracy, and consistency of collected data before analysis.

Why editing is required by fieldworker during data collection
To detect and correct missing, inconsistent, or unclear responses immediately in the field.

Probing – definition
Probing involves asking follow-up questions to get more complete or deeper responses from respondents.

Example of problematic interview question to manager
"Do your staff give full input and behave well?" – It is leading and double-barreled, combining two aspects into one question.

Role of interviewer
To guide the respondent, clarify questions, record accurate responses, and ensure ethical conduct during data collection.

Tips for interviewing
Be neutral, build rapport, listen actively, avoid judgment, and ensure a comfortable environment.

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Principles of interviewing
Ensure clarity, neutrality, consistency, ethical behavior, and active listening.

Duties of field researcher
Collect accurate data, maintain neutrality, protect respondent confidentiality, and report honestly.

Steps in field research
Include selecting the site, gaining access, building rapport, collecting and recording data, and analyzing findings.

What field researchers do
They immerse in natural settings to observe, interact, and collect qualitative or mixed data.

Ethical dilemmas in field research
Include issues of consent, confidentiality, deception, and role conflict during prolonged engagement.

Ethical dilemma faced by interviewer
Includes situations where the interviewer must balance truthful reporting with protecting participant privacy or safety.

Disadvantages of content analysis
It may lack context, be time-consuming, and subject to researcher interpretation bias.

Disadvantages of self-administered or distributed interviews
Low response rate, misunderstanding of questions, and lack of probing opportunities.

Advantages of face-to-face interviews
Allow probing, clarification, better engagement, and higher response quality.

Limitations of non-probability surveys
Results are not generalizable due to sampling bias and lack of randomness.

Disadvantage of content analysis research
It may overlook meaning, nuance, or context behind the content being analyzed.

Historical-comparative research – limitations
Limited data availability, researcher bias, and difficulty in causal inference.

Spontaneity and confession in secondary data – example
Content such as diary entries or online forums may offer spontaneous confessions useful in non-reactive analysis.

How researcher locates evidence in historical-comparative research
Through documents, records, archives, and existing literature that provide context and historical perspective.

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History effect vs mechanical loss in research
History effect refers to outside events influencing results during a study. Mechanical loss refers to equipment failure or technical problems during data collection.

Parts of the table
A table typically includes a title, row and column headings, body (data entries), and source note. Each part helps interpret and structure data for clarity.

Parts of experiments
Key components include:

- **Independent variable** (manipulated factor),
- **Dependent variable** (measured outcome),
- **Control mechanisms** (e.g., control group, randomization),
- **Experimental group** (receives the treatment),
- **Pretest/posttest** (optional), and
- **Treatment/intervention**.

Three types of variables

- **Independent variable**: the cause or input manipulated.
- **Dependent variable**: the effect or outcome measured.
- **Control variable**: held constant to isolate the effect of the independent variable.

Types of probability sampling
Includes:

- Simple random sampling
- Systematic sampling
- Stratified random sampling
- Cluster sampling

Types of non-probability sampling
Includes:

- Convenience sampling
- Purposive sampling
- Quota sampling
- Snowball sampling
- Sequential sampling
- Theoretical sampling

Tools for data collection
May include:

- Questionnaires

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- Interviews
- Observations
- Documents/records
- Focus group discussions (FGDs)
- Scales and tests

Field interview vs survey interview
Field interviews are more flexible, unstructured or semi-structured, and take place in natural settings. Survey interviews are more structured, often conducted via questionnaires, aiming for consistency across many respondents.

Which survey or research is suitable in given conditions
It depends on objectives. For large-scale trends: **survey research**. For in-depth understanding: **qualitative interviews**. For cause-effect: **experimental**. For existing records: **non-reactive research**.

Observation (including example in shopping mall)
Observation is a method of collecting data by watching behavior.
Example: A researcher observes customer buying behavior in a mall without interacting — this is **non-participant, non-reactive observation**.

Explain three important parts of an experiment

1. **Manipulation** of an independent variable
2. **Control** over extraneous variables (e.g., through random assignment)
3. **Measurement** of the dependent variable after treatment

Scenario: "I am satisfied with salary and work environment" – identify mistake
This is a **double-barreled question** — it asks about two things in one sentence, making it unclear which one the respondent is answering.

Define sampling
Sampling is the process of selecting a subset of individuals from a population to study characteristics of the whole population.

Explain difference between population and target population

- **Population**: the entire group a researcher is interested in.
- **Target population**: the specific portion of the population accessible and relevant to the study.

Why defocus mind before field research?
Defocusing helps researchers avoid preconceived notions and observe reality objectively. It encourages openness to unexpected findings in the field setting.

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